SEPARABLE VARIANT OF THE CURVE-BASE NORMALIZED BASE CHANGE: SMOOTHNESS AT GENERIC FIBRE POINTS AFTER NORMALIZATION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 19480-19497 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 19499-19506, the complete author proof (6 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. BORDERLINE for Chief: the proof is an analogy pointer (follows in exactly the same manner that A follows from B) naming two published chapter items as the pattern source; accepted because the pattern's endpoints are explicit in-chapter and the reader can mechanically reconstruct, but flagged for possible rejection if Chief wants pattern-pointers excluded. Statement itself is fully self-contained.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The chapter's curve-case derivation pattern (over-curve lemma), the two theorems (main and separable) of the chapter's normalized-base-change section.

Source reference. Original lemma 0BRQ: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19358–19372. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BRR: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19401–19410. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 09IL: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19282–19289. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1 (Variant with separable extensions over curves)**.** *Let $f : X \to S$ be a flat, finite type morphism of schemes. Assume S is Nagata, integral with function field K, and regular of dimension* 1*. Assume the field extensions $\kappa(\eta)/K$*

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are separable for every generic point η of an irreducible component of X. Then there exists a finite separable extension L/K such that in the diagram

$$\begin{array}{ccccc} Y & \xrightarrow{\nu} & X \times_S T & \longrightarrow & X \\ & \searrow^{g} & \downarrow & & \downarrow f \\ & & T & \longrightarrow & S \end{array}$$

the morphism g is smooth at all generic points of fibres. Here T is the normalization of S in $\mathrm{Spec}(L)$ *and $\nu : Y \to X \times_S T$ is the normalization.*

Proof. This follows from Lemma 0BRR in exactly the same manner that Lemma 0BRQ follows from Theorem 09IL. □